

Manage Storage Like a Pro with LVM

Disk storage space is always at a premium. No matter how much space one has, it always seems to shrink. In this article, the author suggests a solution to this problem with the Linux-based Logical Volume Manager (LVM).



“A hundred gigs ought to be enough for my home partition!” is what you might have said before you typed those fateful commands into *fdisk*. Little did you realise that six months later, you’d be regretting that decision.

When I bought a 1TB external hard drive last year, I thought I was set for five years at the very least. Barely eight months on, I’ve already used up 80 per cent of the disk. And I’m using the rest up at an alarming rate. Trust me, today what might seem like more than enough storage for the rest of your life, will seem like a floppy disk half a year from now. You need to do something about it.

So what *can* you do? You can’t go out and buy an exabyte of storage now. That’s too expensive. What you can do is buy as much as you need now, and just keep adding to it as and when you need more storage. Simple, right? Only if you could take the hassle out of partitioning, filesystems, migration and what not. Well, with Linux, you can.

The logical volume manager or LVM

The big problem with storage seems to be partitions. You never know just how much you’re going to need. You can’t just dump everything into one partition - not only is that very bad housekeeping, but it’s also a good idea to keep system and data separated because if your system goes down, you can

wipe it and start over without worrying about your data.

Partitioning wouldn’t be all that bad if you could only grow, shrink and throw those partitions around whenever you felt like it (or needed to). Well, with LVM, you can.

The big idea behind LVM is that you don’t need to bother if the partitions you store your data in are really physical partitions. All they need to be are logical units - volumes. With LVM, you don’t divide your disk into different partitions; you create logical volumes on them, and then you’re free to do whatever you want with them. If a volume is filling up, just increase its size. If you have no space to do that, just shrink a lesser used partition to make some space.

And LVM really comes into its own when you need to add capacity. Running really low on space? Go out and buy another drive. It can be anything - a 2TB platter drive or a shiny new 256 GB SSD. Or even one of those new fangled Fusion Drives. LVM doesn’t care what it is, as long as it has storage space in it. Connect the drive to your system, tell LVM it’s there, and then grow your volumes, which will now use up the new space available in the new drive.

“If it’s that good, there’s got to be some catch!” you say, deciding that, “Maybe it’s slow.” Well, the magic thing is, it’s not. You might think that layering your filesystem over one additional volume manager might be slow, but it’s not. LVM is built into the Linux kernel, and it uses the same